

FRONT

BACK

Mestinon® Pyridostigmine Bromide

Information for Physicians

Mestinon® Dragées 60 mg
Mestinon® Tablet 10 mg

Mestinon®
Pyridostigmine Bromide
Vagotonic, antimyasthenic

Composition
Tablets 10 mg: round biconvex tablets, white to off white coloured and odourless. Dragées (sugar-coated tablets) 60 mg: round biconvex sugar-coated tablets, orange-white or pale orange coloured.

Properties and Effects
Pyridostigmine, the active ingredient of Mestinon® is a cholinesterase inhibitor. It is characterized by the gentle onset, smooth course, comparatively long duration and gradual decline of its cholinergic action. Pyridostigmine has an antagonistic effect on non-depolarising muscle relaxants only. It has a synergistic effect on depolarizing muscle relaxants.

Pharmacokinetics
Absorption
Pyridostigmine, like other medicines of the same type, is not fully absorbed from the intestinal tract. Bioavailability following oral administration is 3 - 8 %. Considerably higher doses are therefore required for oral as opposed to parenteral administration.

Distribution
Peak plasma concentrations were reached in the fasting state approximately 1.5 - 2 hours after administration of 120 mg pyridostigmine. The increase in active ingredient is delayed when taken with food. The distribution volume averages 1.4 L/kg body weight. Pyridostigmine is not noticeably bound to plasma proteins and does not cross the blood-brain barrier. Blood plasma levels of 20 - 60 ng/ml are required in order to obtain the desired therapeutic effect with myasthenia gravis.

Metabolism
Pyridostigmine is metabolized to 3-hydroxy-N-methylpyridine and other unidentified metabolites.

Elimination
Mean values of around 1.5 hours are given for the elimination half life. This can, however, be prolonged up to three times over in individual cases. Mean plasma clearance in healthy volunteers is given as 0.36 - 0.65 L/kg/h. No confirmed data are available regarding the potential accumulation of unchanged pyridostigmine or active metabolites. Since the dosage must, in any case, be adjusted individually, this point is devoid of practical significance. Pyridostigmine is largely eliminated unchanged (75 - 81%) via the kidneys. One part (18 - 21 %) appears as the 3-hydroxy-N-methyl-pyridine metabolite in the urine. Other unidentified metabolites account for 1 - 4 %.

Kinetics in Specific Clinical Situations
Impaired liver function has no relevant effect on the kinetics of pyridostigmine. The elimination half-life can increase approximately four fold and plasma clearance can fall by up to around one-fifth in the case of age or disease induced kidney failure.

Indications and Usage
Mestinon® is useful in the treatment of myasthenia gravis.

Dosage and Administration
When using Mestinon® it is important to remember that the full effect appears gradually, usually within 15-30 minutes.

Use in Adult
1 - 3 dragées (60 -180 mg) 2 - 4 times daily, or higher doses if required. The total daily dose is usually in the range of 5 - 20 dragées (300 -1200 mg) but doses higher than these may be needed by some patients.

Use in Children
Children under 6 years old should receive an initial dose of 30 mg of Mestinon®. Children 6 - 12 years old should receive 60 mg. Dosage should increase gradually, in increments of 15 - 30 mg daily, until maximum improvement is obtained. Total daily requirements are usually in the range of 30 - 360 mg.

Patients with swallowing difficulties can take tablets broken into small pieces instead of the whole tablets or dragées. In myasthenia gravis, one dose is effective for approximately four hours during the day whilst at night (owing to reduced physical activity), a longer duration of effect of around 6 hours can be expected. It is recommended that the times of administration be chosen so that the maximum effect coincides with the most strenuous physical exertion, e.g. when getting up and at mealtimes.

Special Dosage Instructions
The required dose must be carefully titrated when used in paediatrics. In cases of neonatal myasthenia, generally one treatment with neostigmine (Prostigmin) is preferred. However, if this appears unsuitable due to excessively cholinergic side effects, then Mestinon® can be administered. In these cases, the following serves as a rough guideline: 5 - 10 mg by mouth in tablet form, 30 - 60 minutes before food. Treatment over the eight week if life is required only in extremely rare cases of congenital and familial infantile myasthenia. Pyridostigmine is mainly eliminated unchanged via the kidneys. Lower doses may therefore be indicated in patients presenting with kidney disease. The dose should be adjusted in line with the desired effect.

Restrictions on Use
Contraindications
Mestinon® is contraindicated in mechanical obstruction of the intestinal or urinary tract and known allergy to the preparation.

Precautions
Pyridostigmine is mainly excreted unchanged by the kidney. Therefore, lower doses may be required in patients with renal diseases, and treatment should be based on titration of drug dosage to effect.

Warnings
Although failure of patients to show clinical improvement may reflect under dosage, it can also be indicative of overdose. As is true of all cholinergic drugs, overdosage of Mestinon® may result in cholinergic crisis, a state characterized by increasing muscle weakness which, through involvement of the muscles of respiration, may lead to death. Myasthenic crisis due to an increase in the severity of the disease is also accompanied by extreme muscle weakness, and thus may be difficult to distinguish from cholinergic crisis on a symptomatic basis. Such differentiation is extremely important, since increases in doses of Mestinon® or other drugs of this class in the presence of cholinergic crisis or of a refractory or "Insensitive" state could have a grave consequence. The different diagnosis of the two types of crisis may require the use of Tensilon (edrophonium chloride) as well as clinical judgment. The treatment of the two conditions obviously differs radically. Whereas the presence of myasthenia gravis suggests the need for more intensive anti-cholinergic therapy, the diagnosis of cholinergic crisis, calls for the prompt withdrawal of all drugs of this type. The immediate use of atropine in cholinergic crisis is also recommended. Atropine may abolish or obtund gastrointestinal side effects or other muscarinic reactions: but such use, by masking the signs of overdose, can lead to inadvertent induction of cholinergic crisis.

Pregnancy, Lactation
Reproduction studies in laboratory animals have not indicated any risks to the foetus but no controlled studies have so far been conducted in pregnant women. Therefore, this medication should only be used during pregnancy strictly as directed, with careful dosing and under medical supervision. Since the possibility of pyridostigmine diffusing into breast milk cannot be ruled out, nursing mothers are advised to refrain from breast feeding their babies during treatment.

Undesirable Effects
Like all cholinergic drugs, Mestinon® can have undesirable functional effects on the autonomic nervous system. Muscarine-type side effects can manifest themselves as nausea, vomiting, diarrhea, stomach cramps, increased peristalsis and bronchial secretion, salivation and lacrimation as well as bradycardia and miosis. Nicotine-type side effects consist mainly of muscle spasms, muscle twitches and muscle weakness. Like other bromine-containing drugs, Mestinon® can occasionally cause skin rash, although this generally disappears rapidly once medication has been discontinued. Further use of Mestinon® or other bromine-containing preparations is then contra-indicated.

Interactions
Pyridostigmine antagonizes the action of curare-like non-depolarising muscle relaxants. Atropine counteracts the cholinergic effects of pyridostigmine.

Overdosage
Overdosage with Mestinon® or other cholinesterase inhibitors may lead to cholinergic crisis that manifest themselves in, among other things, pronounced muscle weakness or exacerbated muscle weakness in patients with myasthenia. If such a situation is overlooked, life is endangered due to paralysis of the respiratory muscles. Bradycardia and paradoxically, tachycardia are other possible effects. Counter measures include the immediate withdrawal of Mestinon® or other cholinergics and the slow intravenous administration of 1 - 2 mg atropine sulphate. Depending on the pulse rate, this dose is to be repeated at intervals of two to four hours if required.

Stability
This medicine should not be used after the expiry date (EXP) shown on the pack. For storage remark, please refer to the outer carton.

Packs
Tablets 10 mg: 50
Dragées (sugar-coated tablets) 60 mg: 20,150
* Not all pack sizes are available in all countries

This is a medicament
- A medicament is a product which affects your health, and its consumption contrary to instructions is dangerous for you. Follow strictly the doctor's prescription, the method of use and the instructions of the pharmacist who sold the medicament.
- The doctor and the pharmacist are experts in medicine, its benefits and risks.
- Do not by yourself interrupt the period of treatment prescribed for you.
- Do not repeat the same prescription without consulting your doctor.

Medicament: Keep out of the reach of children

Manufactured by
Labiana Pharmaceuticals S.L.U., Barcelona, Spain

for
A. Menarini Asia-Pacific Holdings Pte. Ltd.
30 Pasir Panjang Road,
#08-32 Mapletree Business City,
Singapore 117440

Date of Revision: March 2022



Final Fold : 3x Horizontal
Fold Size : 135 x 32.5 (mm)

■ Black
IPM No.: 2022MEN-0088-1
Product Name: Mestinon 60mg 150s dragees Insert
Country: MY
Version: 2
Date: 11 March 2022

135 mm

135 mm